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Model Hamiltonian Analysis of Singlet Fission from First Principles

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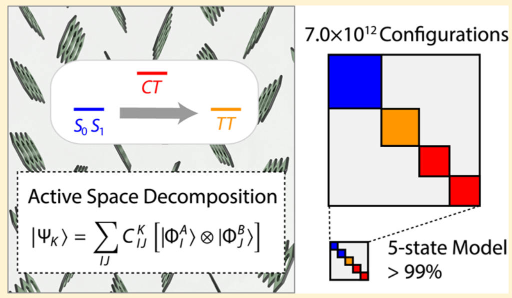
Supporting Information

ABSTRACT: We present an approach to accurately construct the few-state model Hamiltonians for singlet fission processes on the basis of an ab initio electronic structure method tailored to dimer wave functions, called an active space decomposition strategy. In this method, the electronic structure of molecular dimers is expressed in terms of a linear combination of products of monomer states. We apply this method to tetracene and pentacene, using monomer wave functions computed by the restricted active space (RAS) method. Near-exact wave functions are computed for π-electrons of dimers that contain up to 7×10^{12} electronic configurations. Our product ansatz preserves the diabatic picture of the minimal dimer model, allowing us to accurately identify model Hamiltonians. The wave functions obtained from the model Hamiltonians account for more than 99% of the total wave functions. The resulting model Hamiltonians are shown to be converged with respect to all the parameters in the model, and corroborate previously reported coupling strengths.



■ INTRODUCTION

Singlet fission is a molecular process by which one photogenerated singlet exciton splits into two triplet excitons. The produced triplets are entangled such that the overall spin state is singlet, thus making the process spin-allowed and potentially fast,[1,2] in some cases proceeding as quickly as a few hundred femtoseconds.[3] Interest in singlet fission has rapidly expanded, fueled by its potential to significantly improve photovoltaic devices via carrier multiplication.[4] In this regard, it is the molecular analogue of multiple exciton generation. The singlet fission process is

$$S_0 \xrightarrow{\hbar\omega} S_1 \rightarrow TT \rightarrow T_1 + T_1$$

where S_n are singlet states, TT stands for the pair of spin-coupled triplets, and T_1 is an independent triplet. It is also represented schematically in Figure 1.

Rapid singlet fission processes are often observed when excited states are ordered as $E(S_1) \geq E(TT)$ with $E(S_1)$ and $E(TT)$ being the excitation energies of the S_1 and TT states with respect to the ground state. Ignoring the difference

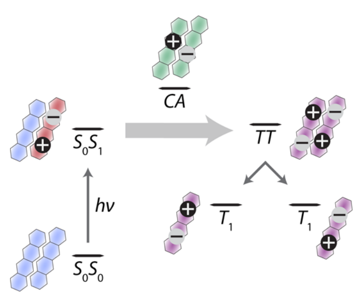


Figure 1. Singlet fission mechanisms.

between $E(TT)$ and $2E(T_1)$ [$E(T_1)$ is the excitation energy of the T_1 state] suggests that it is desirable for a singlet fission chromophore to satisfy the energy matching criterion

$$E(S_1) \geq 2E(T_1) \tag{1}$$

which has been among the few design principles for singlet fission materials to date. This condition is satisfied by pentacene's excited states, in which $E(S_1) - 2E(T_1) \approx 0.1-0.2$ eV, and fission in pentacene occurs on a time scale of 80–200 fs.[1]

Tetracene is, however, an exception. Experiments have estimated the singlet absorption energy and the triplet energy to lie at 2.35 eV[5] and 1.25 eV,[6] respectively, violating the energy matching criterion [$E(S_1) - 2E(T_1) \approx -0.2$ eV]. Nonetheless, triplet yield in tetracene approaches 200%,[7] indicating near quantitative conversion. Chan et al.[6] explained this apparent contradiction with entropic arguments, estimating a critical temperature of $T = 170$ K, above which fission is no longer activated (and thus implying temperature dependence for $T <$ 170 K).[6] However, recent experiments have demonstrated that fission remains temperature independent as low as 10 K.[5,8,9] Burdett et al.[8] and Tayebjee et al.[9] proposed dark intermediates that are lower in energy than S_1 and from which fission commences, although the identities of their proposed states differ.

Computational predictions of singlet fission rates and triplet yields have used few state model quantum dynamics simulations,[10,11] which require a model Hamiltonian as an input. The model dimer Hamiltonian for singlet fission[1] includes the low-lying single-exciton states (S_- and S_+, which are linear combinations of S_0S_1 and S_1S_0), the double-exciton

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state (TT), and charge-transfer states (AC and CA, where A and C refer to anionic and cationic states, respectively):

$$H_{\text{model}} = \begin{pmatrix} H_{S_-,S_-} & H_{S_-,S_+} & H_{S_-,TT} & H_{S_-,CA} & H_{S_-,AC} \\ H_{S_+,S_-} & H_{S_+,S_+} & H_{S_+,TT} & H_{S_+,CA} & H_{S_+,AC} \\ H_{TT,S_-} & H_{TT,S_+} & H_{TT,TT} & H_{TT,CA} & H_{TT,AC} \\ H_{CA,S_-} & H_{CA,S_+} & H_{CA,TT} & H_{CA,CA} & H_{CA,AC} \\ H_{AC,S_-} & H_{AC,S_+} & H_{AC,TT} & H_{AC,CA} & H_{AC,AC} \end{pmatrix} \quad (2)$$

in which $H_{X,Y} = \langle X|\hat{H}|Y\rangle$. From this model Hamiltonian, two mechanisms are apparent. In the direct mechanism, entangled triplets are generated by the direct coupling from S_- (or S_+) to TT. In the other mechanism, the triplet-pair production is mediated by charge-transfer states, AC and CA.

Parameterization of this model Hamiltonian is challenging since it involves one- and two-electron excited states. Standard time-dependent density functional theory (DFT) is unsuitable since it does not describe two-electron excited states properly.[12] Single-reference coupled-cluster approaches can, in principle, describe these states; however, since triple-excitation operators are required to accurately describe two-electron excited states, its applicability is limited to small molecules.[13] Similarly, active-space based methods are capable of accurately simulating both one- and two-electron excited states; however, their factorial scaling with respect to number of π-orbitals makes accurate calculations of chromophore dimers intractable. In addition, conventional electronic structure methods only give adiabatic states, not the diabatic states used in the model; therefore, most electronic structure simulations of singlet fission to date have not been used to parametrize the model dimer Hamiltonians[14−17] (although diabatic states can be found via postprocessing[18] or approximated as combinations of isolated monomer states[19]). As a result, model Hamiltonians have been constructed by qualitative methods that are based on the highest-occupied and lowest-unoccupied molecular orbitals (HOMO and LUMO) of each monomer.[20−23] For instance, Berkelbach et al. constructed model Hamiltonians and used them in Redfield quantum dynamics studies to investigate the mechanism of fission in pentacene.[10,11] Similarly, Renaud et al. combined their computed couplings with a simple kinetic model to predict the singlet fission yield of perylene based chromophores as a function of geometric stacking.[24]

While excitations within the HOMO and LUMO space comprise the dominant contributions to the monomer wave functions, they neglect a significant portion of the exact monomer wave functions. For example, see Figure 2 in which

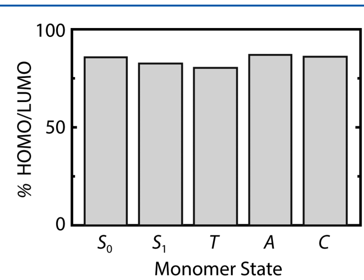

Figure 2. Proportion of each monomer state with HOMO/LUMO character (computed with 6 orbitals in RAS **II**—see Computational Details).

we plot the weights assigned to HOMO/LUMO configurations of several electronic states of tetracene by a more accurate electronic structure method (RAS(6,6,6)[1,1], with all π orbitals correlated, as described in Computational Details). The HOMO/LUMO picture certainly captures the character of the monomer wave functions, but the missing 17−23% precludes quantitative accuracy. In this study, we present a method to compute ab initio model Hamiltonians, on the basis of a product wave function ansatz that preserves the diabatic view of the model Hamiltonians and achieves quantitative accuracy at the same time. Model Hamiltonians for two prototypical singlet fission chromophores, tetracene and pentacene, will be computed and discussed using this method. Note that our model states are strictly speaking quasidiabatic,[25] but we will hereafter refer to them as diabatic as there is no risk of confusion.

■ ELECTRONIC STRUCTURE OF DIMERS

We recently developed an approach to efficiently construct a dimer's wave function from purely monomer calculations. It parametrizes a dimer wave function in terms of monomer states

$$|\Psi_K\rangle = \sum_{IJ} C_{IJ}^K [|\Phi_I^A\rangle \otimes |\Phi_J^B\rangle] \equiv \sum_{IJ} C_{IJ}^K |\Phi_I^A \Phi_J^B\rangle \quad (3)$$

where A and B label monomer units, I and J label monomer states, and K labels dimer states. The monomer wave functions Φ_I^A and Φ_J^B form a basis in which the dimer states are represented. Using monomer states that diagonalize a monomer Hamiltonian, eq 3 converges quickly, as we will show later. These decomposed wave functions approach exact dimer wave functions in two limiting cases: (i) when there is no coupling between the two monomers and (ii) when I and J are complete sets of monomer states. A complete set of monomer states in this context includes all possible charge and spin states. This allows us to guarantee the precision of the direct product ansatz by monitoring the convergence of the dimer Hamiltonian.

This direct product ansatz is convenient for the computation of quantitative model Hamiltonians for singlet fission. In the previous study,[26] we have shown that the dimer Hamiltonian matrix elements between direct product basis

$$H_{IJ,I'J'} = \langle \Phi_I^A \Phi_J^B | \hat{H} | \Phi_{I'}^A \Phi_{J'}^B \rangle \quad (4)$$

can be computed from only monomer transition density matrices without explicitly constructing dimer wave functions. In addition, the direct product structure of the wave function ansatz preserves the interpretive power of the dimer model.

The direct product ansatz requires one to choose the number and type of monomer states used to construct the dimer space. By type, we refer to the charge, q, and spin multiplicity, S, of each monomer. While our implementation can compute matrix elements between arbitrary types of monomer states, for our present purpose, we restrict ourselves to those types that play a central role in singlet fission: singlet, triplet, anionic, and cationic. For each type of monomer state, we compute and store M monomer states with maximum spin projection ($m_s = S$). For convenience, we use the same value of M for all monomer subspaces of both dimers. For $S \neq 0$, the rest of the spin manifold ($m_s \neq S$) is generated by successively applying the spin-lowering operator. Only monomer states for which $q^A + q^B = 0$, $S^A = S^B$ and $m_s^A + m_s^B = 0$ will contribute to a charge neutral spin singlet wave function, where the superscript refers to the monomer. We will show below that the results are sufficiently converged with small M. For further discussion of this ansatz, see ref 26.

■ COMPUTATIONAL DETAILS

Molecular Geometry. Tetracene and pentacene crystals both adopt similar herringbone lattices (see Figure 3). For all geometries considered, we use the experimentally determined crystal structures.[27] There are three distinct nearest-neighbor dimers due to the symmetry of each crystal. These three dimer pairs can be seen from the possible pairings of the three labeled monomers in Figure 3. All other nearest-neighbor pairs are related to these by some combination of the crystals' translational and inversion symmetries. These three dimers are given the notation "dimer α" (monomers 1 and 2), "dimer β" (monomers 1 and 3), and "dimer γ" (monomers 2 and 3). We compute model Hamiltonians for all three tetracene dimers. However, we found dimer α to have the strongest coupling in tetracene, and hence we will limit our discussion to dimer α of tetracene. The results for the other two dimers are collected in the Supporting Information. For pentacene, we compute only the model Hamiltonian for dimer α.

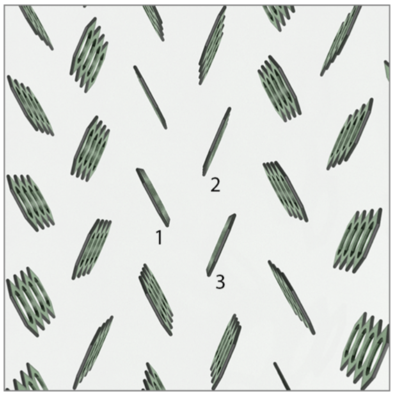

Figure 3. Herringbone lattice of the tetracene crystal. Pentacene adopts a similar lattice (from the above perspective, tetracene and pentacene are indistinguishable).

To mimic the effects of the crystal environment, each dimer is embedded in the middle of a 5 × 5 × 3 crystal supercell, where all other units are represented by atom-centered point charges. Hence, each dimer is surrounded by about 1 nm of crystal in all directions. The point charges were chosen to represent the electrostatic environment resulting from the crystal and were determined as follows. First, a DFT calculation was performed on the dimer and point charges centered at the dimer's nuclei were determined by fitting them to reproduce the computed electrostatic potential. Next, the point charges derived in the previous step are placed in the appropriate lattice positions of the 5 × 5 × 3 supercell. Then another DFT calculation was performed in the presence of these point charges and the resulting electrostatic potential was used to obtain new point charges. Finally, this process was repeated until self-consistency (about 10 iterations). We used the PBE functional[28] with the def2-TZVP basis set.[29] All DFT calculations and electrostatic potential fits were performed using Turbomole.[30]

Choice of Dimer Active Orbitals. The following localization procedure was applied separately to the sets of canonical occupied and virtual orbitals arising from a dimer Hartree–Fock calculation in order to define monomer orbitals. First, the canonical orbitals were localized and assigned to each fragment using the Pipek–Mezey algorithm[31] by maximizing $L = \sum_i [(P_i^A)^2 + (P_i^B)^2]$, where P_i^X is the Löwdin population of orbital i on fragment X. Diagonalizing the Fock matrix within each block of orbitals belonging to the same monomer yields a full set of molecular orbitals that are localized to each monomer, and the resulting orbitals resemble isolated monomer molecular orbitals. Finally, the orbitals that have the greatest overlap with reference active orbitals obtained from minimal-basis calculations on isolated monomers are chosen to be active.

Monomer Wave Function Method. Monomer states are computed using the RAS model,[32] in which the active orbitals are split into three subsets (referred to as **I**, **II**, and **III**) and the configuration space is restricted such that **I** has a minimum occupation number (or a maximum number of holes) and **III** has a maximum occupation, while **II** has no restriction on occupations. A particular RAS model is defined by assigning active orbitals to each subset and choosing the allowed number of holes in **I**, h, and the allowed number of particles in **III**, p. In effect, this model fully correlates the electrons in the RAS **II** space, while allowing select interactions with the RAS **I** and **III** spaces, depending on the chosen values of h and p (e.g., orbital relaxation for $1h$ and $1p$, minimal dynamic correlation for $2h$ and $2p$, etc.). Owing to these restrictions, the number of active orbitals used in RAS can be significantly larger than what is possible for the complete active space model. A RAS model is delineated by the notation RAS(l,m,n)[h,p], where l, m, and n are the number of orbitals in the **I**, **II**, and **III** subsets, respectively, and h and p are the maximum number of allowed holes and particles in the **I** and **III** subsets.

The full suite of π-orbitals in tetracene (18 orbitals) and pentacene (22 orbitals) are thus included in the active space. Among many ways to partition the π-orbitals into the three RAS subsets, we have found from isolated monomer calculations that the RAS(l,m,l)[1,1] model with m = 4,6,8; $l = (N_\pi - m)/2$, and the def2-TZVPP basis set[29] provides a satisfactory balance of accuracy and computational efficiency, where N_π is 18 for tetracene and 22 for pentacene. We will distinguish these RAS models using the number of orbitals in RAS **II**, m.

The space spanned by the product of two monomer wave functions computed by the RAS model corresponds to a more complex occupation-restriction structure than RAS itself: Each monomer can have up to h holes or p particles, and no restrictions are placed on the occupations of the union of both RAS **II** orbital sets. The tetracene product space contains 1.8×10^8, 1.9×10^{10}, and 2.0×10^{12} configurations for 4, 6, and 8 orbitals in RAS **II**, respectively. For pentacene, the spaces contain 4.5×10^8, 5.5×10^{10}, and 6.8×10^{12} configurations. The advantage of our product state ansatz is twofold. First, it enables calculations that are otherwise impossible. Second, it allows us to focus on physically relevant parts of the dimer space. A product of RAS with $1h$ and $1p$ includes orbital relaxation between RAS **I**, **II**, and **III**, while excluding dynamical correlation in RAS **I** and **III**, as in the corresponding monomer calculations. This is not possible with theories in which only the numbers of holes and particles for the total system are specified.

Although the quality of excitation energies from RAS calculations is known to be deficient, relative energies are captured quite well. All computations to form the model

Hamiltonian were performed in the open source BAGEL package.[33]

■ MODEL HAMILTONIANS

We construct the model Hamiltonians by computing model states and evaluating the exact Hamiltonian matrix elements on the basis of those model states. The model states are chosen by diagonalizing the sub-blocks of the dimer Hamiltonian with the desired diabatic character and extracting the relevant low-lying block eigenstates. The sub-blocks are singlet (SS), two-exciton (TT), and charge-transfer (AC and CA, or CT) blocks. There are $(2S_{\mathrm{block}}+1)M^2$ states in each block where S_{block} is the spin eigenvalue of the monomer states (S_{block} = 0, 1, 1/2, 1/2 for the SS, TT, AC, and CA blocks, respectively). Thus, for $M = 16$, the SS, TT, AC, and CA blocks have 256, 768, 512, and 512 dimer states. The first two excited eigenstates of the SS subspace are included in the set of model states and are termed $|S_-\rangle$, and $|S_+\rangle$. Similarly, the lowest eigenstate in the TT, AC, and CA subspaces is termed $|TT\rangle$, $|AC\rangle$, and $|CA\rangle$, respectively. Since singlet fission dynamics typically start from a photoexcited initial state, we omit the ground state of the SS subspace from our discussion. The resulting model states have the proper diabatic characters and are singlet coupled.

First, we numerically show that the model Hamiltonians converge rapidly with M, one of the parameters in our dimer wave function ansatz. Since there is an arbitrary phase associated with each model state, we will focus on the diagonal entries of the model Hamiltonian (e.g., H_{S_+,S_+}) and the magnitude of the coupling matrix elements (e.g., $|H_{S_+,AC}|$). Figures 4 and 5 demonstrate the convergence of the diagonal energies and the magnitudes of the coupling matrix elements of the tetracene and pentacene model Hamiltonians with respect to two important parameters: the number of states in each monomer subspace, M, and the number of orbitals in the RAS **II** space. Figure 4(a1),(a2) shows the convergence of the tetracene model Hamiltonian with respect to M for the RAS(6,6,6)[1,1] model space (i.e., 6 orbitals in RAS **II**) and Figure 5(a1),(a2) shows the corresponding plots for pentacene using the RAS(8,6,8)[1,1] model space. The rapid convergence of the models demonstrates the effectiveness of the direct product ansatz; already by the minimal $M = 2$, the models are essentially converged.

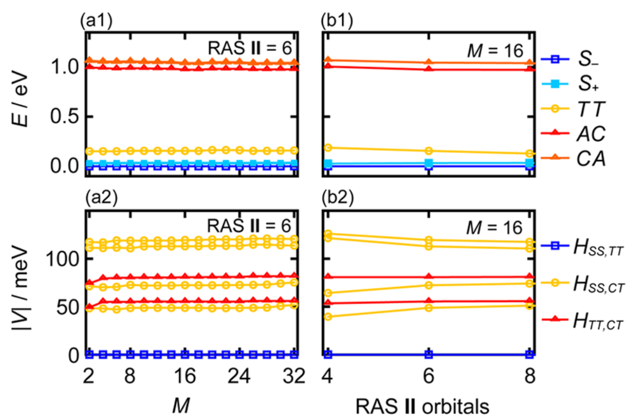

Figure 4. Convergence of the model Hamiltonian for tetracene dimer α with respect to (a) M and (b) number of RAS **II** orbitals per monomer. (a1) and (b1) are those of the diagonal energies, and (a2) and (b2) are those of the magnitude of the couplings. CT refers to either AC or CA.

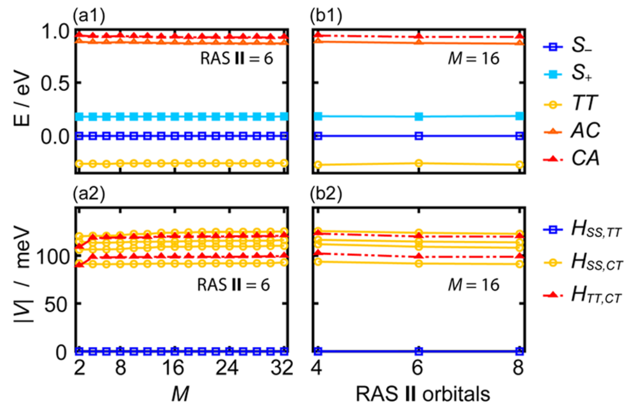

Figure 5. Convergence of the model Hamiltonian for the pentacene dimer with respect to (a) M and (b) number of RAS **II** orbitals per monomer. (a1) and (b1) are those of the diagonal energies, and (a2) and (b2) are those of the magnitude of the couplings. CT refers to either AC or CA.

Next, we show that the model Hamiltonians also converge rapidly with the number of orbitals in RAS **II**, the parameter that controls the accuracy of monomer wave functions. Panels (b1) and (b2) show the effect of an expanding active space on the elements of the model Hamiltonians for the tetracene dimer (Figure 4) and the pentacene dimer (Figure 5). Note that unions of RAS **I**, **II**, and **III** are always the full set of π-orbitals in this study. Both the diagonal and off-diagonal elements of the model Hamiltonians do shift, but only slightly, as the active space is expanded. The apparent larger dependence of $H_{SS,CT}$ to the number of RAS **II** orbitals in Figure 4(b2) is mainly ascribed to the mixing between S_- and S_+, which lie close in energy, and not to errors. This result justifies the use of small active spaces (as small as 4 orbitals in RAS **II** per monomer) for qualitative calculation of the coupling matrix elements. For quantitative simulations, however, it is necessary to use larger active spaces. The convergence patterns for the other tetracene dimers considered are nearly identical, and are collected in the Supporting Information.

From the tests above, we choose $M = 16$ with 6 orbitals in RAS **II** to construct our final model Hamiltonians. We demonstrate the accuracy of the 5-state model Hamiltonian by computing the norm of the adiabatic states projected onto the space orthogonal to the model space, i.e.,

$$\Delta_K = \|(\hat{1} - \hat{P}_{\mathrm{model}})|\Psi_K\rangle\|^2 \tag{5}$$

where $\hat{P}_{\mathrm{model}}$ is the projector onto the model space and Ψ_K is an eigenstate of the full dimer Hamiltonian with the same M. Δ_K is thus the proportion of the dimer adiabatic states that is absent from the model space. Figure 6 shows the values of Δ_K for the lowest four dimer adiabatic states of each dimer. These four states are composed primarily of mixtures of the ground state, the two single-exciton states, and the double-exciton state, although they also contain contributions from the charge-transfer states (AC and CA). All dimer states shown have Δ_K < 0.75% (Δ_K < 0.5% for tetracene), indicating that the electronic structure of the dimer is well captured by the set of diabatic model states (99% or more). Feng et al. argued that the few states model is inappropriate for the tetracene dimer since the diabatic states that they considered (HOMO/LUMO excitations) only accounted for 74−95% of the adiabatic wave

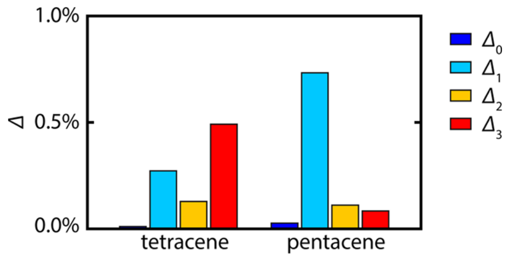

Figure 6. Norm of the adiabatic dimer wave functions projected onto the space orthogonal to the model space (eq 5) for dimer α of tetracene and pentacene.

functions.[16] Our encouraging result should be ascribed to the fact that our basis states are diabatic by construction, facilitating the analysis of adiabatic wave functions in terms of a diabatic basis.

The resulting model Hamiltonian for tetracene dimer α is (in the same order as in eq 2)

$$H^{t}_{\mathrm{model}} = \begin{pmatrix} 0.000 & 0.000 & 0.000 & -0.051 & -0.074 \\ 0.000 & 0.034 & 0.000 & -0.118 & -0.111 \\ 0.000 & 0.000 & 0.128 & -0.081 & 0.056 \\ -0.051 & -0.118 & -0.081 & 0.972 & 0.001 \\ -0.074 & -0.111 & 0.056 & 0.001 & 1.039 \end{pmatrix} \quad (6)$$

in units of eV, where, again, S_- is defined as the zero of energy. Those for other dimers can be found in Supporting Information. Our numerical results support several qualitative features of that reported in previous studies in which the couplings between singlet and charge-transfer states are on the order of 50−120 meV and those between charge-transfer and two-exciton states are on the order of 50−80 meV.[34] In addition, the matrix elements responsible for so-called direct singlet fission, $\langle S_-|\hat{H}|TT\rangle$ and $\langle S_+|\hat{H}|TT\rangle$, are all smaller than 1 meV, thus indicating that the charge-transfer mediated mechanism is likely to dominate in tetracene. The corresponding model Hamiltonian for the pentacene dimer is

$$H^{\mathrm{p}}_{\mathrm{model}} = \begin{pmatrix} 0.000 & 0.000 & -0.000 & 0.091 & -0.114 \\ 0.000 & 0.187 & -0.000 & 0.123 & -0.108 \\ -0.000 & -0.000 & -0.270 & -0.120 & -0.099 \\ 0.091 & 0.123 & -0.120 & 0.866 & -0.002 \\ -0.114 & -0.108 & -0.099 & -0.002 & 0.930 \end{pmatrix} \quad (7)$$

also in units of eV. The diagonal elements of the model Hamiltonian corresponding to the charge-transfer intermediates are lowered and the couplings between singlet and charge-transfer states and those between two-exciton and charge-transfer states are greater in magnitude relative to tetracene, facilitating the charge-transfer mediated mechanism.

For comparison, we also computed model Hamiltonians for all dimers using RAS(0,2,0)[0,0] monomer wave functions which is equivalent to the HOMO/LUMO picture. Interestingly, the magnitudes of the coupling elements change by only 20−25% when using the full π space versus using only the HOMO/LUMO space. The diagonal elements of the model Hamiltonian, by contrast, are much more sensitive to the size of the active space and shift by up to 0.8 eV. These minimal model Hamiltonians are collected in the Supporting Information.

■ CONCLUSIONS

We have presented a novel approach to parametrize, from first-principles, a model Hamiltonian corresponding to the minimal dimer model of the singlet fission process. Our approach is based on the active-space decomposition strategy, recently developed by the authors, which is tailored to molecular dimers. The dimer wave function is expanded as a linear combination of products of monomer states. We use the RAS model to compute monomer wave functions, allowing all the π orbitals to be treated in the active space. The direct product ansatz allows us to not only achieve computational efficiency, but also map first-principles computations to the diabatic picture in the minimal model Hamiltonian. The model Hamiltonians resulting from this method have been shown to converge rapidly with respect to all the parameters in the active-space decomposition method. Furthermore, we validated the diabatic picture of singlet fission in tetracene and pentacene by computing the norm of the adiabatic states projected onto the model space. Therefore, our numerical results (eqs 6 and 7) can be used to gauge dimer model Hamiltonians obtained by more approximate theories. Our approach is readily applicable to other chromophore dimers for singlet fission, paving the way for rational design of such chromophores from first principles. Since the full dimer space is never explicitly constructed, our method also extends the reach of such calculations to any chromophore dimers for which the computation of several monomer states is feasible. In addition, our method is currently being adapted to treat more strongly coupled dimers (such as covalently linked dimers) or molecular aggregates and to include dynamical correlation, which we expect to have the strongest impact on the absolute excitation energies and the weakest impact on coupling magnitudes.

■ ASSOCIATED CONTENT

Supporting Information

The model Hamiltonians and their corresponding convergence data are shown for the other tetracene dimers as well as minimal model Hamiltonians for all dimers considered. In addition, some algorithm details and computational costs are presented. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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